Ayush Maurya

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SUMMARY

Research practitioner with a dual-degree foundation (B.Tech + M.S. by Research) bridging computational methods and human sciences. Work spans agent-based simulation systems for electoral modeling, computer vision models for visual geolocation, and document digitization pipelines combining OCR with LLM-based text recovery. Applies rigorous experimental methodology across every project: hypothesis design, controlled evaluation, and reproducible documentation. Brings the depth of computational social science to industry research.

EDUCATION

IIIT Hyderabad — 2021–Present
B.Tech in Computer Science and M.S. by Research in Computing and Human Sciences
CHD: Computer Science and Humanities Integrated Dual Degree (Offered by the Human Sciences Research Center)

Mary Gardiner's Convent School — 2021
Class XII — 90.1%

City Montessori School — 2019
Class X — 94.2%

EXPERIENCE

SERC Digital Twin — *Environmental IoT*

- Developed a real-time environmental monitoring dashboard using the MERN stack.
- Integrated local sensors for temperature, humidity, water, and air quality monitoring.

TAFEA — *System Design*

- Designed a student client software system using standard SDLC methodologies.
- Integrated LLM APIs to support tailored student-facing features and workflows.

PROJECTS

ElectoralSim — Agent-Based Electoral Simulation

- Published open-source research prototype for vectorized agent-based electoral simulation on PyPI.
- Implemented modular simulation framework supporting multiple electoral systems with documented API.

ML for Local Geoguesser — Visual Geolocation via Deep Learning

- Designed controlled experiments comparing ResNet50 architectures for visual geolocation inference.
- Achieved 92% region accuracy through iterative hypothesis testing and statistical model evaluation.

Highland History Project — OCR + LLM Pipeline

- Built OCR-LLM pipeline digitizing historical documents for Himalayan trade-network research.
- Evaluated multiple LLM configurations for degraded-text recovery with accuracy benchmarks.

Design for Social Innovation (TAFEA) — AI-Assisted Education Platform

- Developed AI-assisted platform enabling curriculum design for resource-constrained classrooms.
- Conducted human-centered research and evaluated prototype effectiveness through user studies.

Age Prediction Model — Transfer Learning vs Ensemble

- Conducted comparative study of transfer learning versus ensemble methods for age prediction.
- Formulated experimental hypotheses, implemented in PyTorch, and documented statistical findings.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL, HTML/CSS
Technologies: Machine Learning, PyTorch, Linux, System Design, Systems Programming
UI/UX and Design: Wireframing, Prototyping, User Stories, User Research, Interaction Design

HONOURS

UGEE Rank: 111 out of 40,000+ candidates

ACTIVITIES

Bass guitarist in college band and lawn tennis player.